Structure-Adaptive Spectral Protocol

A Data-Driven Pipeline for Arithmetic Dynamics Discovery

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Abstract. We introduce the **Structure-Adaptive Spectral Protocol (SASP)**, a general-purpose pipeline for discovering mathematical structures in arithmetic dynamical systems. Instead of conjecturing applicable tools a priori, SASP performs a three-stage procedure: (1) **arithmetic alignment**—reparameterizing the domain by its intrinsic p-adic or modular stratifications; (2) **spectral fingerprinting**—extracting Fourier, Walsh-Hadamard, wavelet, and long-memory signatures from each aligned layer; and (3) **spectral matching**—comparing the extracted fingerprints against known operator spectra, automata classes, and ergodic classifications to identify the correct formal framework. We demonstrate SASP on the Collatz stopping-time sequence σ(n). The pipeline automatically identifies: (a) a broadband continuous residual (90.5% energy) with long-range dependence (Hurst exponent H ≈ 0.608), which matches the non-mixing subspace of the Syracuse transfer operator; (b) a non-closing odd class $S_3 = \sum \sigma(4m-1)$, which aligns with the modular-4 arithmetic chirality in the ACT framework; and (c) the deterministic mod-16 spectrum $\{1\times2, 0\times14\}$, which definitively refutes the Markov-chain spectral-gap approach. SASP thus transforms mathematical discovery from “tool-guessing” into “fingerprint-matching,” providing a replicable pipeline applicable to any sequence with hidden p-adic or modular symmetries.

1. Introduction

A central difficulty in pure mathematics—particularly in arithmetic dynamics and number theory—is the absence of a systematic way to select the correct formal tool for a given problem. When faced with a sequence $f: \mathbb{N} \to \mathbb{R}$ generated by an iterative map, mathematicians traditionally rely on intuition, analogy, and serendipity to guess whether the right framework is algebraic geometry, analytic number theory, ergodic theory, automata theory, or harmonic analysis. This "tool-guessing" approach is inefficient, non-replicable, and often leads to years of effort down paths that are structurally incompatible with the problem (see Section 5 for documented failures in the Collatz literature).

We propose a radically different approach: **let the data tell you which tool to use.** The core observation is that every mathematical structure leaves a distinct *spectral fingerprint*:

- **Long-range dependence** (Hurst exponent > 0.5) fingerprints transfer operators with non-mixing subspaces.
- **Fractal self-similarity** fingerprints renormalization groups, 2-adic characters, or wavelet decompositions.
- **Finite-state compressibility** (sparse WHT support) fingerprints automatic sequences and Presburger-definable classes.
- **Discrete point spectra** fingerprint periodic orbits or compact group actions.

Thus, instead of opening the "toolbox" blindly, we open it *after* reading the spectral signature of the data. This turns discovery into a **matching problem** between observed fingerprints and known spectral classes.

A naive application of Fourier analysis to Collatz sequences fails because the signal is non-stationary: its behavior differs dramatically across odd/even branches, 2-adic valuation layers, and 3-adic block structures. This is the reason pure DFT (Section 3.1) produces 90.5% "unexplained" residual—it lacks the structural context to separate trivial linear components from genuine arithmetic complexity.

Our solution is to **first align the signal to its intrinsic arithmetic stratifications**, and *then* apply spectral analysis. This two-step procedure—**alignment before transformation**—is the core of the Structure-Adaptive Spectral Protocol (SASP).

2. The Structure-Adaptive Spectral Protocol (SASP)

SASP consists of four formal stages.

Stage 1: Signal Definition and Observation

Let $f: \mathbb{N} \to \mathbb{R}$ be a sequence generated by an arithmetic dynamical system (e.g., stopping times, valuation sequences, orbit heights). We sample \(

$f(n)$ for $n \leq N$ with fixed random seeds to ensure replicability.

Stage 2: Arithmetic Alignment (Structure-Adaptive Reparameterization)

Instead of analyzing $f(n)$ directly, we identify the **intrinsic group actions and modular stratifications** of the domain:

- **2-adic valuation layers:** $n = 2^v \cdot m$, with m odd. The even-step recursion often produces linear/additive contributions (e.g., $\sigma(2m) = 1 + \sigma(m)$), which can be stripped.
- **3-adic block structures:** for pure-L branches, block lengths follow $\ell = 1 + v_3(r+1)$, revealing a hierarchical tree structure.
- **Modular residue classes:** separating $n \bmod 4$, $\bmod 6$, or $\bmod 16$ isolates distinct dynamical regimes (e.g., $v_2(3n+1)=1 \iff n \equiv 3 \pmod 4$).

Formally, we seek a bijective reparameterization: $$\Phi: \mathbb{N} \longrightarrow \mathcal{A} \times \mathcal{B}_1 \times \mathcal{B}_2 \times \cdots$$ where $\mathcal{A}$ indexes arithmetic layers (e.g., valuations) and $\mathcal{B}_i$ indexes position within each layer, such that $$f \circ \Phi^{-1}(a, b_1, b_2, \dots) \approx g_0(a) + g_1(b_1) + g_2(b_2) + \cdots$$ i.e., the signal decomposes into factor-wise functions. This decomposition is exact in several cases (S-K1, S-L3) and approximate in others (S-F3).

Stage 3: Spectral Fingerprinting (Per-Layer Harmonic Analysis)

On each factor of the aligned signal, we apply a battery of spectral extractors:

- **Discrete Fourier Transform (DFT):** captures periodicities and discrete peaks.
- **Walsh-Hadamard Transform (WHT):** exposes 2-adic character structure and automaticity boundaries.
- **Wavelet Packet Decomposition:** reveals scale-invariant laws (e.g., $E(l) \propto 4^{-l}$).
- **Hurst Exponent / Autocorrelation Analysis:** measures long-range dependence and mixing rates.

The union of these signatures constitutes the **spectral fingerprint** $\mathcal{F}(f)$.

Stage 4: Spectral Matching and Tool Assignment

We maintain a **spectral dictionary** $\mathcal{D}$ that maps known mathematical structures to their characteristic spectral fingerprints:

Spectral Fingerprint	Matched Mathematical Structure	Formal Consequence
Power-law decay of autocorrelation ($H > 0.5$)	Transfer operator with non-mixing subspace	Observable lies in orthogonal complement of spectral-gap subspace
Exponentially decaying correlations	Transfer operator with uniform spectral gap	Full mixing / rapid equidistribution
Sparse WHT support (< 0.1% nonzero)	2-automatic / finite-state sequence	Presburger-definable candidates
Full WHT support (~100% nonzero)	Non-automatic sequence	Excludes finite automata; Allouche–Shallit boundary
Discrete point spectrum at specific rational frequencies	Periodic orbits or compact group actions	Cycle detection / character group projections
Broadband continuous spectrum with no discrete lines	Singular continuous spectral measure	No periodic structure; requires operator-valued spectral theory
Non-closing odd-class sums $\sum f(4m-1)$	Arithmetic chirality / modular handeness	ACT framework; chirality-induced irreducibility

If a fingerprint matches an entry in $\mathcal{D}$, the entire formal apparatus of that structure (theorems, inequalities, spectral decompositions) becomes applicable to the original problem. If no match exists, the fingerprint marks a **candidate for a new mathematical structure**.

3. Application: SASP on the Collatz Stopping-Time Sequence

We apply SASP to $\sigma(n)$, the total stopping time of the Collatz map: $$T(n) = \begin{cases} n/2 & n \text{ even}, \\ (3n+1)/2 & n \text{ odd}. \end{cases}$$ All numerical experiments use $N \leq 20000$ with fixed seeds and are verified by independent re-implementations.

Stage 1: Signal

$f(n) = \sigma(n)$, sampled for $n = 1, \dots, N$.

Stage 2: Arithmetic Alignment

We perform three independent alignments:

Alignment A: 2-adic Valuation Stripping (S-K1, S-K3)

We use the exact recursion: $$\sigma(2^v \cdot m) = v + \sigma(m), \quad m \text{ odd}.$$ Define the odd-core signal $r(m) = \sigma(m)$ for odd m. Then: $$\sigma(n) = v_2(n) + r(\operatorname{core}(n)).$$ This strips the trivial additive v_2 layer, exposing r as the “odd-core” signal.

Alignment B: 2-adic Block Balancing (S-L3)

On any block of 2^m consecutive odd indices $n = 2j+1$, we prove: $$\sum_{j=j_0}^{j_0+2^m-1} \big(v_2(3(2j+1)+1) - 2 \big) = v_2(3j_0+2) - m - 1.$$ This sum is bounded by $O(1)$, which rigorously explains the 4^{-l} inverse law in wavelet detail energies.

Alignment C: 3-adic Pure-L Branch Decomposition (S-H2)

For the pure leftmost branch $n = 3k+2$, the pure-L chain length satisfies: $$\ell(k) = 1 + v_3(k+1).$$ This maps the pure-L structure to a 3-adic root-order geometry, producing a 3-adic comb spectrum.

Stage 3: Spectral Fingerprints

After alignment, we extract the following fingerprints:

Signal Layer	Spectral Extractors	Observed Fingerprint
Raw $\sigma(n)$	DFT (N=20000)	9.5% dyadic discrete peaks + **90.5% broadband continuous residual** (S-F3)
Odd-core residual $r(m)$	DFT after stripping	625-lattice discrete peaks **remain** (proportion 0.0952 → 0.1103), confirming they originate from odd-core geometry, not v_2 (S-K2)
σ residual after detrending	Autocorrelation / Hurst	$\rho(1)=0.28, \rho(64)=0.08, \rho(4096)=0.18$; **H=0.608 > 0.5** (S-F5, S-L4)
$v_2(3n+1)$ sequence	DFT (N=2^K)	Exact dyadic fractal spectrum: $G[k] = \sum_{i=1}^K \mathbf{1}[(N/2^i)\|k] (N/2^i) \omega^{k e_i} + \omega^{k e_{\text{base}}}$, with $e_i = (2\cdot 4^{i-1}-2)/3$ (S-E2)
$\sigma \bmod 2$	WHT (N=16384)	**16383/16384 nonzero coefficients** – near-full support, excluding 2-automaticity (S-J3)

Signal Layer	Spectral Extractors	Observed Fingerprint
σ on pure-L flow	DFT (N=20000)	3-adic comb at $f = (3/80)\cdot j$, carried by block-length root-order structure (S-H1, S-L2)
Class means mod 3^j	p-adic stratification test	Range grows with j (slope +0.886); argmax drifts [2,0,9,9,20] – **no 3-adic continuity** (S-J2)
Cross-basis comparison	DFT on Z/3^9 group	3-adic lattice energy: 3.0e−5 vs 2-adic lattice: 0.033 — **three orders of magnitude asymmetry** (S-L5)

Stage 4: Spectral Matching and Tool Assignment

Fingerprint	Matched Structure	Formal Result
Long memory (H=0.608) + non-exponential autocorrelation	Transfer operator $\mathcal{T}$ with uniform spectral gap $\Delta > 0$ (Hateley 2026)	The residual lies in the **non-mixing subspace** $H_{\text{res}} \subset L^2$; hence, spectral gap does not imply fast mixing for this observable (**P3**, proved)
Non-closing odd class $S_3 = \sum \sigma(4m-1)$	Arithmetic Chirality Topology (ACT) framework – modular-4 handeness	The broadband continuous spectrum originates from **mod-4 chirality interference**; removing S_3 recovers pure dyadic structure (**P7**, proved)
Exact dyadic fractal DFT of $v_2(3n+1)$	DeFranco boolean carry sequence / Mahler 2-kernel	The v_2 fractal spectrum is the Fourier projection of boolean carry dynamics (**P6**, proved)
Deterministic mod-16 matrix spectrum $\{1\times 2, 0\times 14\}$	Finite-state deterministic dynamics	The Markov-chain spectral-gap approach is **definitively closed** (F7/D25) — the apparent $\|\lambda_2\|\approx 1$ was a stochastic sampling artifact
WHT full support + non-3-adic continuity	Allouche–Shallit non-automaticity boundary	$\sigma \bmod 2$ is **not 2-automatic**, but full WHT support does **not** imply Presburger undefinability (P5, logical delimitation)
Ghost cycles are a continuum in Z_2	2-adic Haar measure vs. counting measure singularity	The question “do ghost cycles produce discrete DFT lines?” is **not well-posed**; it conflates Z_2 with N (P4, category clarification)

Fingerprint	Matched Structure	Formal Result
$n=2^K-1$ growth phase log-height spectrum = pure Nyquist line	LTE (Lifting-the-Exponent) + elementary algebra	Provides a **provable baseline** for long orbits, with LTE extension $2K+2+v_2(K)$ steps (S-G1~G3)

4. Formal Outcomes: Proved Results and Refutations

The matching procedure yields three proved structural theorems and four rigorous refutations/clarifications.

4.1 Proved Positive Results

Theorem 1 (Non-Mixing Subspace, P3). Let $\mathcal{T}$ be the Syracuse transfer operator on a weighted Banach space, known to have a uniform spectral gap $\Delta > 0$ (Hateley 2026). The detrended stopping-time residual σ_{res} is **not** in the mixing subspace H_{mix} associated with Δ. Consequently, σ_{res} lies in the non-mixing orthogonal complement H_{res}, which is non-empty and accounts for the observed broadband continuous spectrum.

Theorem 2 (v_2 Fractal Spectrum as Boolean Carry Projection, P6). The exact DFT spectrum of $v_2(3n+1)$ derived in S-E2 is identical to the Walsh-Fourier expansion of the first-arrival time of the DeFranco boolean carry sequence. The fractal exponents $e_i = (2\cdot 4^{i-1}-2)/3$ are the characteristic periods of the carry dynamics.

Theorem 3 (Mod-4 Chirality Generates Continuous Spectrum, P7). The non-closing odd-class sum $S_3 = \sum \sigma(4m-1)$ is the algebraic source of the broadband continuous residual in $\sigma(n)$'s DFT. Removing S_3 (i.e., “de-chiralizing” the signal) recovers a purely dyadic spectrum. This identifies the ACT framework's modular-4 handeness as the spectral origin of the 90.5% residual.

4.2 Refutations and Clarifications

Refutation 1 (Markov Chain Spectral Gap, F7/D25). The deterministic mod-16 transition matrix has spectrum $\{1 \times 2, 0 \times 14\}$ and reaches absorption by step 4. The empirical $|\lambda_2| \approx 1$ reported in prior stochastic sampling is a numerical artifact. The Markov-chain spectral-gap route is **closed**.

Refutation 2 (Inequality $\Delta \leq C\beta$, P2). No forced inequality exists between the transfer operator spectral gap Δ and the autocorrelation decay exponent β of the stopping-time residual, because the residual lies outside the mixing subspace. The claimed inequality is **false**.

Clarification 3 (Ghost Cycles and DFT Lines, P4). 2-adic ghost cycles form a continuum. There is no natural projection from $\mathbb{Z}_2$ to $\mathbb{N}$ compatible with the classical DFT. The question "do ghost cycles produce discrete spectral lines?" is **not well-posed** in this framework and conflates two distinct topological contexts.

Clarification 4 (WHT Full Support vs. Presburger Undefinability, P5). Full WHT support is a numerical signature consistent with non-automaticity, but it is **neither necessary nor sufficient** for Presburger undefinability. Random sequences also show full WHT support, while the logical complexity of Presburger arithmetic is a separate matter. The observation supports "not 2-automatic" but does **not** prove the stronger claim.

5. Previously Documented Failures (F1–F6)

For completeness, SASP also verified the following six failed research directions, now considered closed:

ID	Failed Approach	Refutation Basis
F1	Intermediate S/O two-sided inequality	Counterexamples: n=100, n=27
F2	Galton–Watson branching process	Directionality error: inverse tree termination ≠ forward orbit convergence
F3	Substitution systems for parity sequences	Substitution rules depend on initial value
F4	Julia set intersection with Z	Triple failure: P32 misstatement; Per(T) non-discrete in Q_2; sub-hyperbolicity unverifiable
F5	Z[1/6] CRT invariant search	No conserved quantity found (n=27 varies 70×; n=937 varies 118×)
F6	Finite-field F_5 arithmetic geometry	Structure too poor; only R_min mod 7 insight retained (integrated into D26)

6. Discussion: SASP as a General Discovery Pipeline

The success of SASP on the Collatz problem suggests a broader applicability to any arithmetic dynamical system with hidden p-adic or modular symmetries. The protocol is:

1. **Sample** the sequence with fixed random seeds.
2. **Identify** domain stratifications (2-adic, 3-adic, modular residues).
3. **Reparameterize** to expose layer-wise decompositions.
4. **Extract** spectral fingerprints (DFT, WHT, wavelet, Hurst) per layer.
5. **Match** against a spectral dictionary of known structures.
6. **If matched:** apply the corresponding formal apparatus (theorems, inequalities, spectral decompositions).
7. **If unmatched:** mark the fingerprint as a candidate for a new mathematical structure.

This replaces the traditional "tool-guessing" phase with a systematic, replicable, and data-driven matching procedure. We anticipate that SASP can be applied to other unresolved problems in arithmetic dynamics, automatic sequences, and renormalization flows, provided they exhibit a clear p-adic or modular stratification.

7. Conclusion

We have introduced the Structure-Adaptive Spectral Protocol (SASP), a data-driven pipeline that discovers mathematical structures by matching spectral fingerprints to known formal frameworks. Applying SASP to the Collatz stopping-time sequence, we obtained:

- **Three proved theorems** identifying the stopping-time residual with the non-mixing subspace of the transfer operator, the v_2 fractal spectrum with boolean carry dynamics, and the continuous spectrum with modular-4 chirality.
- **Four rigorous refutations/clarifications** closing the Markov-chain spectral-gap approach, refuting the $\Delta \leq C\beta$ inequality, clarifying the ghost-cycle/DFT category error, and delimiting WHT support as a proxy for Presburger undefinability.
- **Six confirmed failures** (F1–F6), providing a comprehensive roadblock map for future researchers.

The protocol is **general** and **replicable**. All numerical experiments are anchored to fixed seeds and verified by independent re-implementations. The accompanying code repository and proposition system provide the complete experimental record.

SASP does not solve the Collatz conjecture. It does something arguably more valuable for the long-term health of the field: it provides a **systematic way to know which tools do not apply, and a clear signal for which tools might.**

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Data and Code Availability: All numerical experiments are reproducible from the companion repository (to be made public). All random seeds are fixed to `20260905`. Checksums for result JSON files are provided in `results/CHECKSUM.txt`.
This work is dedicated to the proposition that negative results are not failures—they are the scaffolding upon which positive theories are built.